

A NON-ASYMPTOTIC BOUND WITH THE SHARP POWER OF c FOR THE EIGENVALUE DEFECT $1 - \lambda_n(c)$ OF TIME AND FREQUENCY LIMITING

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ABSTRACT. Let $\lambda_0(c) > \lambda_1(c) > \dots$ be the eigenvalues of the time and frequency limiting operator with kernel $\sin c(x - y)/(\pi(x - y))$ on $[-1, 1]$. Fuchs' asymptotic formula $1 - \lambda_n(c) \sim 4\sqrt{\pi} 8^n c^{n+1/2} e^{-2c}/n!$ describes the defect of the top eigenvalues as $c \rightarrow \infty$. Rigorous non-asymptotic bounds in the literature give the decay rate e^{-c} . We prove that for $0 \leq n \leq 4$ and every $c \geq 10\pi$,

$$1 - \lambda_n(c) \leq C_n c^{n+1/2} e^{-2c},$$

with explicit constants C_n . The exponent e^{-2c} and the power $c^{n+1/2}$ are those of Fuchs' formula. The proof combines Slepian's identity $\partial_c \log \lambda_n = 2\psi_n(1)^2/c$ with a sharp upper bound for the mass of the band-limited extension of ψ_n outside $[-1, 1]$: this mass is at most $(\psi_n(1)^2/c)(2/\pi)\mathbf{K}(\chi_n/c^2)(1 + 15/c^2)$, where $\mathbf{K}$ is the complete elliptic integral. The bound comes from a Bessel-type Liouville transformation at the regular singular point $t = 1$, in which the first-order error cancels. All constants are established with ball arithmetic. In a companion paper, the case $n = 4$ is used to bound the bottom of windowed Weil quadratic forms, which are related to the Riemann Hypothesis through Weil's criterion.

Status. Proved in this paper, with explicit constants established by ball arithmetic (Arb). Not formalized. This paper has not yet been reviewed by independent human experts.

1. INTRODUCTION

For $c > 0$ let Q_c be the operator on $L^2(-1, 1)$ with kernel $\sin c(x - y)/(\pi(x - y))$. Its eigenvalues $1 > \lambda_0(c) > \lambda_1(c) > \dots > 0$ are simple, and the eigenfunctions $\psi_n = \psi_{n,c}$ are the prolate spheroidal wave functions (PSWFs) [11, 6]. We normalize them by $\|\psi_n\|_{L^2(-1,1)} = 1$. For fixed n , the defect $1 - \lambda_n(c)$ tends to 0 as $c \rightarrow \infty$, and Fuchs [5] (see also [10]) gave the asymptotic formula

$$(1.1) \quad 1 - \lambda_n(c) \sim \frac{4\sqrt{\pi} 8^n}{n!} c^{n+1/2} e^{-2c} \quad (c \rightarrow \infty).$$

Non-asymptotic bounds for $\lambda_n(c)$ are available in many regimes [8, 9, 2, 3, 1]. For the defect of the top eigenvalues, Bonami, Jaming and Karoui [1, Thm. 3.2] proved

$$\lambda_n(c) \geq 1 - \frac{7}{\sqrt{c}} \frac{(2c)^n}{n!} e^{-c} \quad (0 \leq n \leq c/2.7),$$

whose exponential rate is e^{-c} , not e^{-2c} . We did not find a rigorous non-asymptotic bound with the rate e^{-2c} in the literature.

Theorem 1. *For $n \in \{0, 1, 2, 3, 4\}$ and every $c \geq 10\pi$,*

$$1 - \lambda_n(c) \leq C_n c^{n+1/2} e^{-2c},$$

with the constants of Table 1.

The key estimate concerns the band-limited extension $\tilde{\psi}_n$ of ψ_n to $\mathbb{R}$ (Section 2). Its mass outside $[-1, 1]$ is $T_n := \int_{|t|>1} \tilde{\psi}_n^2 = (1 - \lambda_n)/\lambda_n$.

n	0	1	2	3	4
C_n (Theorem 1)	18.34	155.0	703.4	2340	6621
Fuchs' constant $4\sqrt{\pi}8^n/n!$	7.09	56.7	227	605	1210

TABLE 1. The constants C_n are rounded up; Fuchs' constants are rounded to three digits. The ratios $C_n/(4\sqrt{\pi}8^n/n!)$ are 2.59, 2.73, 3.10, 3.87, 5.48. Each ratio is the product of the factor $e^{\beta_n(c_0)/c_0}$ from the integration in Section 5 and the ratio of $U_n e^{2c_0} c_0^{-n-1/2}$ to Fuchs' constant at c_0 (for $n = 4$: 7.18×0.762). Lemma 5.1 is sharp to 4×10^{-8} .

Theorem 2. *Let $0 \leq n \leq 4$, $c \geq 10\pi$ and $s = \chi_n(c)/c^2$, where χ_n is the Sturm–Liouville eigenvalue (2.1). Then*

$$T_n \leq \frac{\psi_n(1)^2}{c} \cdot \frac{2}{\pi} \mathbf{K}(s) \cdot \left(1 + \frac{15}{c^2}\right), \quad \mathbf{K}(s) = \int_1^\infty \frac{dt}{\sqrt{(t^2 - 1)(t^2 - s)}}.$$

Theorem 2 is sharp to relative order c^{-2} : numerically the ratio of the two sides is 0.985 at $c = 10\pi$ and 0.9925 at $c = 14\pi$ for every $n \leq 4$ (Table 2). The absence of a $1/c$ term in the correction factor is what gives the sharp power: combined with Slepian's identity (Lemma 2.1), it yields $\partial_c \log(1 - \lambda_n) \leq -2 + (2n + 1)/(2c) + O(c^{-2})$, whose integral produces exactly $c^{n+1/2}e^{-2c}$.

In the companion paper [7], Theorem 1 with $n = 4$ is used to bound the bottom of windowed Weil quadratic forms.

Method.

- On $t > 1$ the extension $\tilde{\psi}_n$ solves the radial prolate equation, which has a regular singular point at $t = 1$. A Bessel-type Liouville transformation turns it into Bessel's equation of order 0 plus a perturbation Δ with $c^2\Delta$ bounded uniformly (Section 3).
- The leading term J_0 gives the main term $\mathbf{K}(s)/(\pi c)$ through the identity $\int_0^x y J_0^2 = \frac{x^2}{2}(J_0^2 + J_1^2)$.
- The first-order error of the Volterra iteration is a priori of relative size $1/c$. It cancels to order c^{-2} because the primitive $\int_0^x y J_0 Y_0 = \frac{x^2}{2}(J_0 Y_0 + J_1 Y_1)$ is bounded (Section 4).

2. PRELIMINARIES

Let $\mathcal{F}_c f(x) = \int_{-1}^1 e^{icxy} f(y) dy$. Then $Q_c = \frac{c}{2\pi} \mathcal{F}_c^* \mathcal{F}_c$, and $\mathcal{F}_c \psi_n = \mu_n \psi_n$ with $\lambda_n = \frac{c}{2\pi} |\mu_n|^2$. The functions ψ_n are also the eigenfunctions of

$$(2.1) \quad L_c = -\frac{d}{dx}(1 - x^2)\frac{d}{dx} + c^2 x^2, \quad L_c \psi_n = \chi_n \psi_n,$$

with $\chi_0 < \chi_1 < \dots$. The parity of ψ_n is $(-1)^n$.

Extension. Put $\tilde{\psi}_n(t) = \mu_n^{-1} \mathcal{F}_c \psi_n(t)$ for $t \in \mathbb{R}$. It is entire, it equals ψ_n on $[-1, 1]$, and it satisfies $L_c \tilde{\psi}_n = \chi_n \tilde{\psi}_n$ on $\mathbb{R}$. By Plancherel, $\|\tilde{\psi}_n\|_{L^2(\mathbb{R})}^2 = 1/\lambda_n$, hence $T_n = (1 - \lambda_n)/\lambda_n$. On $t > 1$, with $p = t^2 - 1$ and $q = c^2 t^2 - \chi_n$,

$$(2.2) \quad (p\tilde{\psi}_n)' + q\tilde{\psi}_n = 0.$$

Lemma 2.1 (Slepian's identity). $\partial_c \lambda_n(c) = 2\lambda_n(c)\psi_n(1)^2/c$.

Proof. The eigenvalues are simple and $c \mapsto Q_c$ is a real-analytic family of trace-class operators. By Hellmann–Feynman, $\partial_c \lambda_n = \langle (\partial_c Q_c) \psi_n, \psi_n \rangle$. Since ∂_c of the kernel is $\cos c(x - y)/\pi$ and ψ_n is real,

$$\partial_c \lambda_n = \frac{1}{\pi} \left| \int_{-1}^1 e^{icx} \psi_n(x) dx \right|^2 = \frac{1}{\pi} |\mu_n|^2 \psi_n(1)^2 = \frac{2\lambda_n}{c} \psi_n(1)^2. \quad \square$$

The identity is classical; see [5, 12, 3]. Consequently

$$(2.3) \quad \partial_c \log(1 - \lambda_n) = -\frac{2\psi_n(1)^2}{cT_n}.$$

Lemma 2.2. *For $0 \leq n \leq 4$ and $c \geq 10\pi$, $s = \chi_n(c)/c^2 < 0.3$.*

Proof. By Hellmann–Feynman, $\partial_c \chi_n = 2c \langle x^2 \psi_n, \psi_n \rangle \leq 2\chi_n/c$, since $\chi_n \geq c^2 \langle x^2 \psi_n, \psi_n \rangle$. Hence χ_n/c^2 is non-increasing, and it suffices to take $c = 10\pi$. Since $\chi_n \leq \chi_4$ for $n \leq 4$, it suffices to bound χ_4 , the third even eigenvalue. The matrix of L_c in the even normalized Legendre polynomials is tridiagonal. Rayleigh–Ritz, together with a Sturm-sequence count in ball arithmetic, shows that it has three eigenvalues below $0.3c^2$ at $c = 10\pi$. $\square$

Lemma 2.3. *For $0 \leq n \leq 4$ and $c \geq 10\pi$, $\chi_n \leq (2n+1)c/(1 - \tau_n(c))$, where $\tau_n(c) = \sum_j \int_{|u| > \sqrt{c}} \tilde{H}_j(u)^2 du$. The sum runs over $j \leq n$ with $j \equiv n \pmod{2}$, and $\tilde{H}_j$ are the normalized Hermite functions. At $c = 10\pi$, $\tau_n \leq 3.6 \times 10^{-9}$, and τ_n decreases in c .*

Proof. Use min–max in the parity sector of ψ_n , with trial functions $c^{1/4} \tilde{H}_j(\sqrt{c}x)$ restricted to $[-1, 1]$. The natural boundary condition of L_c requires no cutoff. The form satisfies $\langle L_c f, f \rangle \leq \int_{\mathbb{R}} (f'^2 + c^2 x^2 f^2) \leq (2n+1)c \sum a_j^2$, and $\|f\|_{L^2(-1,1)}^2 \geq (1 - \tau_n) \sum a_j^2$. The tails are bounded by $\int_X^\infty u^m e^{-u^2} du \leq X^{m-1} e^{-X^2}$ for $X^2 \geq m$. $\square$

3. A BESSEL-TYPE LIOUVILLE TRANSFORMATION

Fix $n \leq 4$, $c \geq 10\pi$ and $s = \chi_n/c^2 \in [0, 0.3]$. For $t > 1$ put

$$\xi(t) = \int_1^t \sqrt{q/p}, \quad P = \sqrt{pq}, \quad \tilde{\psi}_n = (\xi/P)^{1/2} W.$$

Then (2.2) becomes

$$(3.1) \quad (\xi W')' + \xi W = \xi \Delta W, \quad \Delta = \frac{(P^{1/2})''}{P^{1/2}} + \frac{1}{4\xi^2} \quad ({}' = d/d\xi).$$

Since $\xi = c\Xi(t, s)$ with Ξ independent of c , the function $F := c^2 \Delta$ depends on (t, s) only. As $t \rightarrow 1^+$ we have $P/\xi \rightarrow 1$, so $W(0) = \psi_n(1)$, and W is the solution regular at $\xi = 0$.

Lemma 3.1. *Put $a = 1 - s$, $y = \sqrt{t^2 - 1}$ and $r = y^2 + a$. Then $F = C + L$, where*

$$C = \frac{1}{4r} + \frac{3 - 6s}{4r^2} + \frac{5sa}{4r^3},$$

and $L \geq 0$ is given by an explicit integral formula. F is positive and decreasing in t , with $F(1, s) = (3 - s)/(3a^2)$. Consequently, for $s \in [0, 0.3]$:

$$\sup_{t>1} F = \text{TV}_t F \leq D_0 := 1.837, \quad \int_1^\infty F d\Xi \leq D_1 := 1.370.$$

Proof. Setup. Let $v = y^2 + 1$, $g(y) = \sqrt{r/v}$ and $\Xi(y) = \int_0^y g$, so that $\Xi'(t) dt = g(y) dy$, $\xi = c\Xi$ and $P = cy\sqrt{r}$. Put $A(y) = \Xi(y)/y = \int_0^1 g(yz) dz$.

Decomposition. Direct differentiation in (3.1) gives $F = B + 1/(4\Xi^2)$, where $B = (l' + l^2 - lg'/g)/g^2$ and $l = 1/(2y) + y/(2r)$. The apparent singularity cancels, and one obtains the exact identity $F = C + L$ with

$$L = \frac{A^{-2} - g^{-2}}{4y^2}.$$

Since $g'(w) = sw/[g(w)(1+w^2)^2]$, integration by parts gives $g - A = y^{-1} \int_0^y wg'(w) dw = y^2\beta$, where

$$\beta(y) = s \int_0^1 \frac{z^2 dz}{g(yz)(1+y^2z^2)^2}.$$

Hence

$$L = \frac{\beta}{4} \left[\frac{1}{A^2 g} + \frac{1}{A g^2} \right] \geq 0.$$

Monotonicity.

- The functions g and A are non-decreasing, and β is non-increasing. So L is non-increasing.
- With $u = y^2$, $-4r^4 \partial_u C = u^2 + (8 - 14s)u + 7 - 5s - 2s^2 > 0$ for $s \leq 0.3$. So C is decreasing.
- Therefore F is positive and non-increasing in t .

Endpoint value. The series $g = \sqrt{a}[1 + sy^2/(2a) + O(y^4)]$, $A = \sqrt{a}[1 + sy^2/(6a) + O(y^4)]$ and $\beta(0) = s/(3\sqrt{a})$ give $F(1, s) = (3 - s)/(3a^2)$. This value increases in s , and at $s = 0.3$ it equals $90/49 < 1.837$. Since F is positive and non-increasing, $\sup F$ and $\text{TV}_t F$ both equal this value.

The integral. Since $F \geq 0$ and $g \leq 1$, we have $\int F d\Xi \leq \int_0^\infty (C + L) dy$.

- The substitution $y = \sqrt{a} \tan \theta$ gives $\int_0^\infty C dy = \pi(20 - 17s)/(64a^{3/2})$.
- Since $A, g \geq \sqrt{a}$, Tonelli gives $\int_0^\infty L dy \leq s\pi/(16a^2)$.

Both terms increase in s . At $s = 0.3$ their sum is $1.36906 \dots < 1.370$, computed in Arb. $\square$

Lemma 3.2 (Volterra form). *Let $M(x)^2 = J_0(x)^2 + Y_0(x)^2$. Then $W/\psi_n(1) = J_0 + \varepsilon_1 + \varepsilon_2$, where ε_1 is the first Picard iterate of*

$$\varepsilon(\xi) = \frac{\pi}{2} \int_0^\xi \eta [J_0(\eta)Y_0(\xi) - J_0(\xi)Y_0(\eta)] \Delta(\eta) (J_0(\eta) + \varepsilon(\eta)) d\eta.$$

Moreover $|\varepsilon_1| \leq \Phi M$ and $|\varepsilon_2| \leq (e^\Phi - 1 - \Phi)M$, where $\Phi \leq \int_0^\infty |\Delta| d\xi \leq D_1/c$.

Proof. The Wronskian of J_0, Y_0 is $2/(\pi\xi)$, which gives the Volterra equation. The kernel is at most $\frac{\pi}{2}\eta M(\xi)M(\eta)$ by Cauchy–Schwarz. Nicholson’s formula [4, 10.9.30] gives $\frac{\pi}{2}xM(x)^2 \leq 1$. Picard iteration then gives the bounds. $\square$

4. PROOF OF THEOREM 2

Let $q(\xi) = c^2(t(\xi)^2 - s)$ and $\omega = -(1/q)' \geq 0$. From $dt = (P/q) d\xi$,

$$T_n = 2 \int_0^\infty \xi W^2 \frac{d\xi}{q}, \quad \int_0^\infty \frac{d\xi}{q} = \frac{\mathbf{K}(s)}{c}.$$

Put $G(x) = \frac{x^2}{2}(J_0^2 + J_1^2) = \int_0^x \eta J_0^2$, $H(x) = \frac{x^2}{2}(J_0 Y_0 + J_1 Y_1) = \int_0^x \eta J_0 Y_0$ and $R = G - x/\pi$. Then

$$\frac{T_n}{2\psi_n(1)^2} = \underbrace{\int \xi J_0^2/q}_{\text{main}} + 2 \underbrace{\int \xi J_0 \varepsilon_1/q + R_2}_X.$$

Lemma 4.1. (BL1) $\frac{\pi}{2}x(J_0^2 + Y_0^2) \leq 1$ for $x > 0$.

(BL2) $\sup_{x>0} |H(x)| \leq h_0 := 0.162$.

(BL3) $|\int_0^\infty R \omega| \leq 1.2 c^{-3}$ for $c \geq 10\pi$, $s \in [0, 0.3]$.

We use the Hankel expansions with explicit remainders. Let $\theta = x - \pi/4$. For $x > 0$,

$$(4.1) \quad H_0^{(1)} = \sqrt{\frac{2}{\pi x}} e^{i\theta} \left(1 - \frac{i}{8x} + e_0\right), \quad H_1^{(1)} = \sqrt{\frac{2}{\pi x}} e^{i(\theta - \pi/2)} \left(1 + \frac{3i}{8x} + e_1\right),$$

with $|e_0| \leq \frac{9}{128}x^{-2}$ and $|e_1| \leq \frac{15}{128}x^{-2}$. This is [4, 10.17(iii)] with $\ell = 2$. It also follows from the integral representation

$$H_\nu^{(1)}(x) = \sqrt{\frac{2}{\pi x}} e^{i(x-\nu\pi/2-\pi/4)} \frac{1}{\Gamma(\alpha+1)} \int_0^\infty e^{-t} t^\alpha \left(1 + \frac{it}{2x}\right)^\alpha dt, \quad \alpha = \nu - \frac{1}{2},$$

and Taylor's theorem with integral remainder.

Proof. (BL1). Nicholson's formula [4, 10.9.30], with $u = 2x \sinh t$, gives

$$x(J_0^2 + Y_0^2) = \frac{4}{\pi^2} \int_0^\infty \frac{K_0(u) du}{\sqrt{1 + u^2/(4x^2)}}.$$

Since $K_0 > 0$ and $\int_0^\infty K_0 = \pi/2$ [4, 10.32.9], the right-hand side is at most $2/\pi$. We have also checked (BL1) independently in ball arithmetic: by Taylor enclosures on 65,520 cells covering $[1/64, 64]$, by Frobenius series on $(0, 1/64]$, and by (4.1) with $\ell = 3$ on $[64, \infty)$.

(BL2). We have $H' = xJ_0Y_0$, so (BL1) gives $|H'| \leq 1/\pi$. The bound is proved separately on three ranges.

- On $[1/64, 64]$: Arb values of H at the cell midpoints, together with $|H'| \leq 1/\pi$, give $|H| < 0.1600$.
- On $(0, 1/64]$: the Frobenius series of J_0, Y_0, J_1, Y_1 give $|H| < 0.001$.
- On $[64, \infty)$: since $JY = \text{Im}((J + iY)^2)/2$, (4.1) gives

$$|H| \leq \frac{1}{2\pi} \left[1 + \frac{1}{8x} + \frac{2(b_0d_0 + b_1d_1)}{x} + \frac{d_0^2 + d_1^2}{x^3} \right],$$

where $b_0 = 1 + 1/(8x)$, $b_1 = 1 + 3/(8x)$, $d_0 = 9/128$ and $d_1 = 15/128$. At $x = 64$ the bracket gives $0.16041 < 0.162$, and the bound decreases in x .

(BL3). For $x \geq 1$ put $E(x) = R(x) + \cos(2x)/(2\pi)$.

- By (4.1), $|E(x)| \leq A_E/x$ for $x \geq 1$, with $A_E = 0.2037 < 1/4$.
- For $0 < x < 1$, the series of J_0 and J_1 give $|E| < 2$.
- In the variable y , $\omega = 2y\sqrt{v}/(c^3r^{5/2})$. The sign of ω_y is that of $a + (2a - 4)y^2 - 3y^4$, so ω is unimodal with $\text{TV } \omega = 2 \max \omega \leq 9/(4\sqrt{3}a^2c^3)$.

Now split $\int R\omega$ into three parts.

- **Cosine term.** Integration by parts against $\sin(2x)/2$ gives $|\frac{1}{2\pi} \int \cos(2x)\omega| \leq \text{TV } \omega/(4\pi)$.
- **E on $(0, 1)$.** Since $\omega(x) \leq 2x/(a^3c^4)$ there, $\int_0^1 |E|\omega \leq 2/(a^3c^4)$.
- **E on $[1, \infty)$.** $\int_1^\infty |E|\omega \leq \frac{1}{4} \int \omega/x dx \leq \pi/(8a^2c^3)$.

With $a \geq 0.7$ and $c \geq 10\pi$, the total is at most $1.198c^{-3}$. $\square$

Remark 4.2. The primitive of R is not bounded. The Hankel expansion gives $R(x) = -\cos(2x)/(2\pi) + (1 - \sin 2x)/(8\pi x) + O(x^{-2})$, so $\int_1^x R = (\log x)/(8\pi) + O(1)$. This is why (BL3) needs the decomposition above.

Proof of Theorem 2. • **Main term.** By parts, $\int \xi J_0^2/q = \int G\omega = \mathbf{K}(s)/(\pi c) + \int R\omega$.

- **Cross term.** Swapping the order of integration,

$$X = \pi \int \eta \Delta J_0 [J_0 A - Y_0 B] d\eta, \quad A = \int_\eta^\infty \xi J_0 Y_0/q, \quad B = \int_\eta^\infty \xi J_0^2/q.$$

By (BL2), $|A| \leq 2h_0/q$. Integrating $\eta J_0 Y_0 = H'$ by parts against ΔB gives $|X| \leq 2\pi h_0 D_0 B(0)/c^2 + 4h_0 D_1/(c^3 a)$ (here $\text{TV } F = \sup F \leq D_0$).

- **Quadratic remainder.** $|R_2| \leq [(e^\Phi - 1)^2 + 2(e^\Phi - 1 - \Phi)] \cdot \frac{2}{\pi} \mathbf{K}(s)/c$, by (BL1).

Relative to $\mathbf{K}(s)/(\pi c)$, and using $\mathbf{K} \geq \pi/2$ and $a \geq 0.7$, the four contributions are at most 2.40, 2.54, 1.88 and 8.02 (times c^{-2}). Their sum is at most $15c^{-2}$. $\square$

5. PROOF OF THEOREM 1

Lemma 5.1 (starting values). *At $c_0 = 10\pi$, $1 - \lambda_n(c_0) \leq U_n$, where*

$$\begin{aligned} U_0 &= 2.0207289 \times 10^{-26}, & U_1 &= 4.9533689 \times 10^{-24}, & U_2 &= 5.9183604 \times 10^{-22}, \\ U_3 &= 4.5903630 \times 10^{-20}, & U_4 &= 2.5967761 \times 10^{-18}. \end{aligned}$$

Proof. • **Trial space.** Fix polynomials $f_0, \dots, f_4$ of degree 126 or 127, with exact rational coefficients, close to $\psi_0, \dots, \psi_4$. By min-max applied to $V_n = \text{span}\{f_0, \dots, f_n\}$,

$$1 - \lambda_n \leq \max_{f \in V_n} \frac{\|f\|^2 - \langle Q_c f, f \rangle}{\|f\|^2}.$$

- **Matrices.** The Gram matrix G follows from the Legendre moments. The matrix of Q_c is replaced by that of the truncated kernel K_N obtained from $\sin z/z = \sum_{k \leq N} (-1)^k z^{2k}/(2k+1)!$ with $N = 160$ (degree 320 in z), integrated termwise. The alternating remainder contributes at most $\varepsilon = 20(20\pi)^{322}/323! < 3 \times 10^{-92}$ to every Rayleigh quotient.
- **Bounds.** A lower bound g_n for the smallest eigenvalue of G and an upper bound R_n for the largest eigenvalue of $G - Q$ follow from row sums. Hence $1 - \lambda_n \leq R_n/g_n + \varepsilon$.

All entries are computed in ball arithmetic at 512 bits. Independent high-precision Legendre-Galerkin computations give $1 - \lambda_n(c_0)$ within the factor $1 + 4 \times 10^{-8}$ of U_n . $\square$

Proof of Theorem 1. By (2.3) and Theorem 2,

$$\partial_c \log(1 - \lambda_n) \leq -\frac{2}{(1+A)(1+B)} \leq -2 + 2A + 2B,$$

where $A = a_n/4 + \frac{9}{64}a_n^2/(1 - a_n)$, $a_n = (2n+1)/(c(1 - \tau_n)) \geq s$ (Lemma 2.3), and $B = 15/c^2$. Here we used $\frac{2}{\pi}\mathbf{K}(s) \leq 1 + s/4 + \frac{9}{64}s^2/(1 - s)$. This equals $-2 + (2n+1)/(2c) + \beta_n(c)/c^2$, where

$$\beta_n(c) = 30 + \frac{(2n+1)c\tau_n}{2(1 - \tau_n)} + \frac{9(2n+1)^2}{32(1 - \tau_n)^2(1 - a_n)}.$$

Here $\tau_n(c)$ is bounded by the explicit Gaussian tail of Lemma 2.3, which is of the form e^{-c} times a polynomial; that bound times c decreases for $c \geq 10\pi$, and a_n decreases. Hence β_n is bounded by a decreasing function of c . Integrating from $c_0 = 10\pi$ gives

$$1 - \lambda_n(c) \leq U_n e^{2c_0} c_0^{-(n+1/2)} e^{\beta_n(c_0)/c_0} \cdot c^{n+1/2} e^{-2c}.$$

Ball arithmetic gives the constants of Table 1. $\square$

6. NUMERICAL ILLUSTRATION

	$n = 0$	1	2	3	4
$1 - \lambda_n(10\pi)$	2.02×10^{-26}	4.95×10^{-24}	5.92×10^{-22}	4.59×10^{-20}	2.60×10^{-18}
ratio in Thm 2, $c = 10\pi$	0.98537	0.98541	0.98545	0.98550	0.98556
ratio in Thm 2, $c = 14\pi$	0.99248	0.99249	0.99251	0.99252	0.99254

TABLE 2. Defects and the ratio $T_n/(\text{right-hand side of Theorem 2})$, computed from Legendre expansions at 200 digits (not certified).

7. REMARKS

- (1) **General n .** The proof uses $n \leq 4$ only through $s \leq 0.3$ and the starting value. For larger n , the same argument works for $c \geq c_0(n)$ once $s \leq s_{\max} < 1$ is ensured, after recomputing D_0 , D_1 and the Bessel constants on $[0, s_{\max}]$.
- (2) **Constants.** Apart from the starting value, the constants lose the factor e^{β_n/c_0} (at most about 7.2 for $n = 4$).

REPRODUCIBILITY

All computer-assisted steps are reproducible from the public repository <https://github.com/moriryota/prolate-weil-bounds>, archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions; this paper corresponds to release v1.0.1). It contains every script with its raw output, the exact rational coefficients of the trial polynomials of Lemma 5.1 together with the certified matrices and bounds g_n , R_n , the interval certificates for (BL1) and (BL2), the Sturm count of Lemma 2.2, and a table matching each constant to its script. All constants are computed in ball arithmetic (Arb, via python-flint, 200–512 bits) with directed rounding asserted in the scripts.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude, Google Gemini and OpenAI GPT. Under the direction of the author, these systems drafted the proofs, implemented and ran the computations, reviewed one another's drafts adversarially, searched the literature, and drafted this manuscript. The author chose the problem and the strategy, assigned and reviewed the tasks, and takes full responsibility for the content. The AI systems are not authors of this paper.

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